COURSE LEVEL PROJECT BASED LEARNING

Real-Time Bus Tracking System

Java Programming Application

A comprehensive object-oriented solution for live public transport tracking, route management, and arrival estimations.

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Team Members

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Course Learning Objectives

Objective	Statement
1	**Define** the basics of Object-Oriented Programming concepts in Java.
2	**Discuss** the use of inheritance, interfaces, packages, and exception handling.
3	**Apply** event handling and multithreading in GUI-based Java applications.
4	**Evaluate** the effectiveness of object-oriented design principles in solving real-world problems.
5	**Compare** the use of interfaces and abstract classes to identify suitable software contexts.
6	**Design** GUI-based applications using AWT/Swing components with event-driven programming.

Course Outcomes (COs)

CO Code	Outcome Statement
CO1	**Identify** real-world transit problems and analyze requirements using Java OOP principles.
CO2	**Design** modular and scalable Java solutions using classes, inheritance, interfaces, and exception handling.
CO3	**Develop** a Java application integrating GUI (AWT/Swing), multithreading, collections, and database operations.
CO4	**Evaluate** functionality, usability, and performance while demonstrating effective teamwork and communication.

Problem Statement & Proposed Solution

Problem Statement

- Commuters face long wait times due to unpredictable public bus schedules and lack of real-time visibility.
- Manual scheduling leads to overcrowding, inaccurate Estimated Times of Arrival (ETAs), and passenger frustration.
- Traditional static transit charts fail to account for traffic delays, route changes, or emergency disruptions.

Proposed Solution

- Develop a Java-based Real-Time Bus Tracking System utilizing Object-Oriented Programming and Swing GUI.
- Simulate live GPS updates using Java Multithreading to provide continuous bus location tracking on dynamic maps.
- Provide passengers with live ETA alerts, route lookup, delay notifications, and bus capacity status.

Project Activities & Milestones

Phase	Activity Description	Milestone Target
PA1 - PA2	Project planning, transit requirements analysis, and Swing UI/UX mockup design.	**M1 - M2** Scope & Wireframe
PA3 - PA4	Java backend setup, class hierarchy creation, data structure selection, and GUI build.	**M3 - M4** Architecture & Front-End
PA5 - PA6	GPS multithreading integration, route calculation logic, data stream processing, and integration.	**M5** Live Core Features
PA7 - PA8	Exception handling, concurrency debugging, performance optimization, and deployment.	**M6** Testing & Stability
PA9 - PA10	Final project review, technical documentation, and post-launch enhancement planning.	**M7** Final Delivery

System Architecture

1. Data Acquisition Layer:

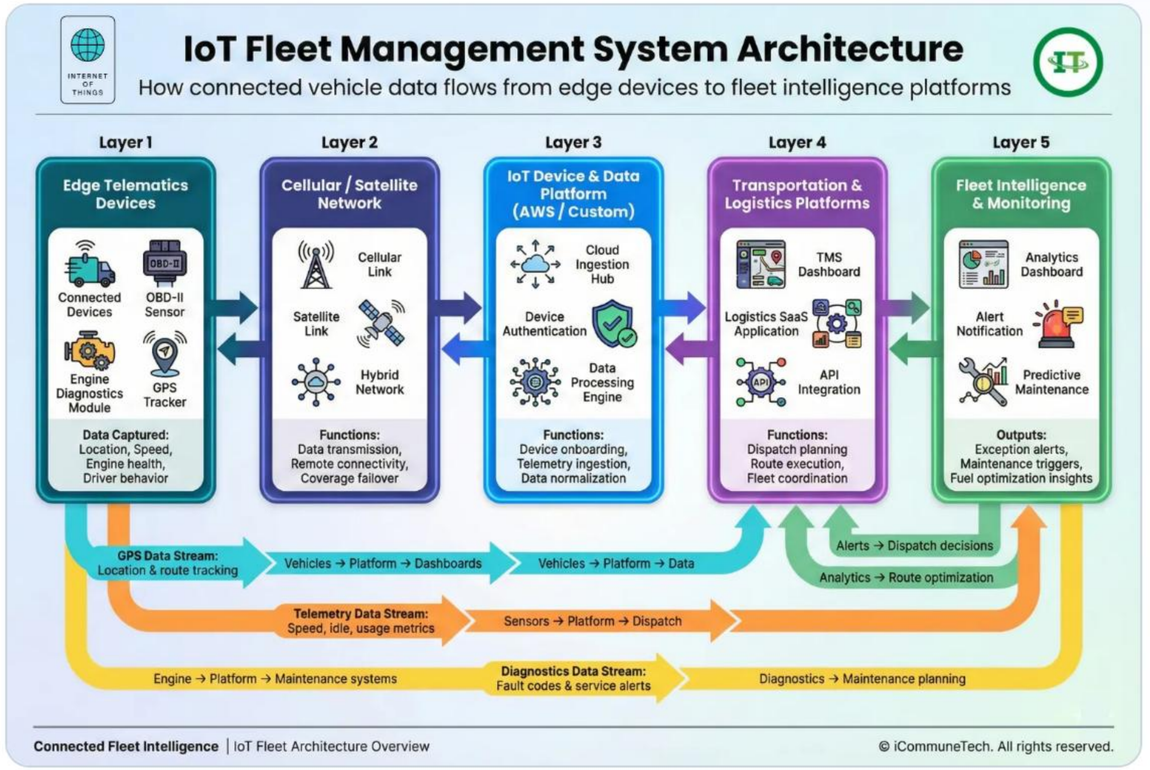
Receives simulated GPS coordinates from bus sensors via multithreaded sockets.

2. Business Logic Layer:

Java core processes ETA calculation, route distance, speed metrics, and stop updates.

3. Presentation Layer:

Event-driven Swing GUI renders interactive map overlays, bus schedules, and passenger notifications.



Core Modules & Java OOP Design

Vehicle & Route Module

Uses Object Encapsulation to store bus IDs, route stop coordinates, driver profiles, and live speed readings.

Multithreaded Tracker

Leverages Java Threads (`Runnable`) to fetch background location feeds without freezing the Swing UI thread.

Interactive Dashboard

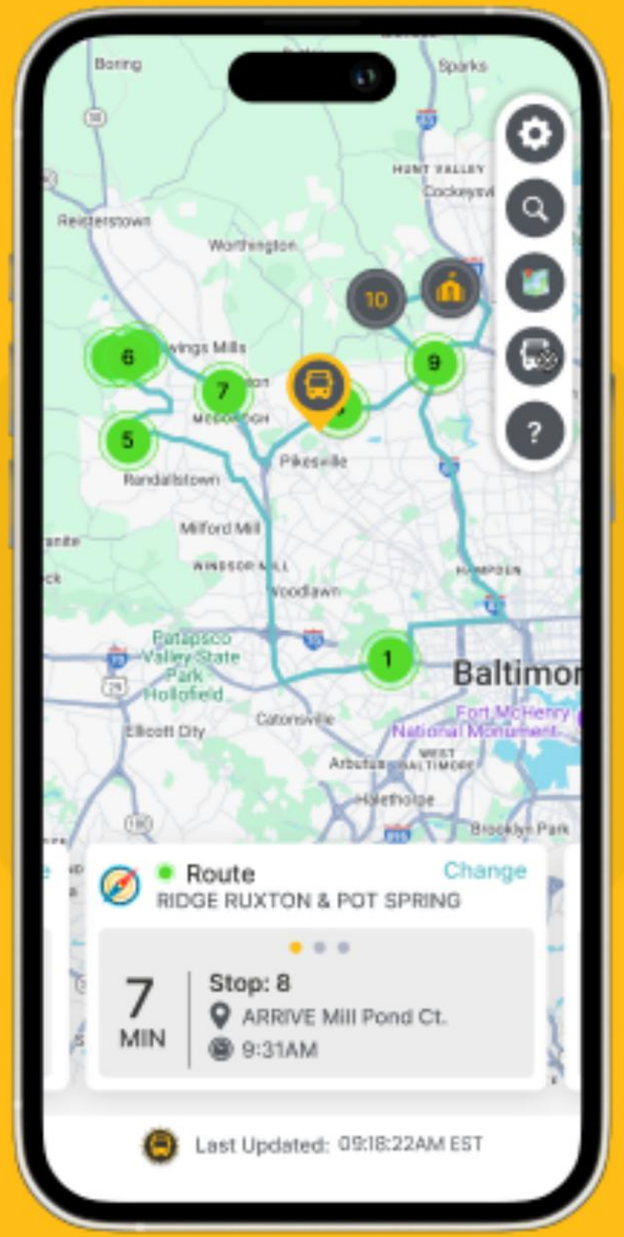
Polymorphic event handlers (`ActionListener`, `MouseListener`) process search inputs and display dynamic ETAs.

Passenger Live Tracking Interface

Intuitive Commuter Dashboard

The Swing-based passenger GUI displays a real-time interactive map with animated bus icons, stop lists, and calculated arrival countdowns.

Features quick route searches, nearby bus stop identification, delay alerts, and seat availability indicators for enhanced commuting convenience.



Key Metrics & System Benefits

35%

Reduction in Commuter Wait Time

Smart Transportation Impact

By supplying live telemetry and automated schedule adjustments, the system optimizes transit efficiency across city transit routes.

- Accurate ETA calculations within a ±1 minute margin of error.
- Reduced station overcrowding through passenger arrival predictions.
- Optimized bus fleet distribution based on real-time crowd demand.

Smart City Fleet Integration

- **IoT GPS Connectivity:** Seamless communication between transit bus telemetry devices and central Java server.
- **Low Latency Updates:** Refresh rates under 2 seconds ensure pinpoint accurate bus location tracking.
- **Robust Error Handling:** Custom exception handlers gracefully manage signal loss and network drops.



Questions & Discussion

Thank You for Your Time!

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Image Sources



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